

Opera Numerorum – M9-All GRH Certification

$X_0(N)$ Curves: Bost–Connes VALOR Table

Battle Plan v1.6 David Fox May 2026

Parent M7 SHA: 5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcfb7ebe3c9

Bost–Connes constant: $C(S_4) = 11.42214869$ Filter: $g \leq 32$, margin > 0 VALOR formula: $\lfloor (C(S_4) - 2\sqrt{g}) \times 10^4 \rfloor$

Rows with CM = 1 are highlighted.

	N	g	$2\sqrt{g}$	C_{S_4}	margin	VALOR	CM	Cert_SHA (first 16)	parent_M7
	11	1	2.00000000	11.42214869	9.42214869	94221	0	4b72e7f19a7a6ca0...	5b80b84d...
	14	1	2.00000000	11.42214869	9.42214869	94221	0	01c4d729f1b35db9...	5b80b84d...
	15	1	2.00000000	11.42214869	9.42214869	94221	0	bcd1da6b9fc4e0f5...	5b80b84d...
	17	1	2.00000000	11.42214869	9.42214869	94221	0	29eb38218a653cf9...	5b80b84d...
	19	1	2.00000000	11.42214869	9.42214869	94221	0	1af61195a58a8700...	5b80b84d...
	20	1	2.00000000	11.42214869	9.42214869	94221	0	124432d5ba50f4aa...	5b80b84d...
	21	1	2.00000000	11.42214869	9.42214869	94221	0	c86e83a9a19fc39a...	5b80b84d...
	22	2	2.82842712	11.42214869	8.59372157	85937	0	955c35e780c7e1fd...	5b80b84d...
	23	2	2.82842712	11.42214869	8.59372157	85937	0	1f4c47513d2ea457...	5b80b84d...
	24	1	2.00000000	11.42214869	9.42214869	94221	0	5250a9aabc75d066...	5b80b84d...
	26	2	2.82842712	11.42214869	8.59372157	85937	0	94bc26dc070ed8e9...	5b80b84d...
cmpink!20	27	1	2.00000000	11.42214869	9.42214869	94221	1	8292c9c947b04093...	5b80b84d...
	28	2	2.82842712	11.42214869	8.59372157	85937	0	843839e909385b01...	5b80b84d...
	29	2	2.82842712	11.42214869	8.59372157	85937	0	3d0594c4af693487...	5b80b84d...
	30	3	3.46410162	11.42214869	7.95804707	79580	0	eef467f0d9a436aa...	5b80b84d...
	31	2	2.82842712	11.42214869	8.59372157	85937	0	5d63a553adb66f52...	5b80b84d...
cmpink!20	32	1	2.00000000	11.42214869	9.42214869	94221	1	0bd9ec3fa4526c28...	5b80b84d...
	33	3	3.46410162	11.42214869	7.95804707	79580	0	5ebb4c2b2db0c017...	5b80b84d...
	34	3	3.46410162	11.42214869	7.95804707	79580	0	ee2df7443db93e95...	5b80b84d...

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M9-All GRH Table (continued)

	N	g	$2\sqrt{g}$	C_{S_4}	margin	VALOR	CM	Cert_SHA (first 16)	parent_M7
cmpink!20	35	3	3.46410162	11.42214869	7.95804707	79580	0	1fbeeabd303b0886...	5b80b84d...
	36	1	2.00000000	11.42214869	9.42214869	94221	1	074533a9e415e0cc...	5b80b84d...
	37	2	2.82842712	11.42214869	8.59372157	85937	0	89eec149fe99492b...	5b80b84d...
	38	4	4.00000000	11.42214869	7.42214869	74221	0	31aa36b2f792e97b...	5b80b84d...
	39	3	3.46410162	11.42214869	7.95804707	79580	0	7d607b2f71691c22...	5b80b84d...
	40	3	3.46410162	11.42214869	7.95804707	79580	0	ecc249cea1cd6948...	5b80b84d...
	41	3	3.46410162	11.42214869	7.95804707	79580	0	2ceb5737b80b08c4...	5b80b84d...
	42	5	4.47213595	11.42214869	6.95001274	69500	0	132a7abed5679449...	5b80b84d...
	43	3	3.46410162	11.42214869	7.95804707	79580	0	4d5412fee349e569...	5b80b84d...
	44	4	4.00000000	11.42214869	7.42214869	74221	0	1d3af239f8224681...	5b80b84d...
cmpink!20	45	3	3.46410162	11.42214869	7.95804707	79580	0	4e3bae2732a618f1...	5b80b84d...
	46	5	4.47213595	11.42214869	6.95001274	69500	0	0f25d882a0839133...	5b80b84d...
	47	4	4.00000000	11.42214869	7.42214869	74221	0	855ecdeb522febf3...	5b80b84d...
	48	3	3.46410162	11.42214869	7.95804707	79580	0	1ca9cd32f70c10de...	5b80b84d...
	49	1	2.00000000	11.42214869	9.42214869	94221	1	4c6f9b4b75ab8993...	5b80b84d...
	50	2	2.82842712	11.42214869	8.59372157	85937	0	b85e24548b798ea9...	5b80b84d...
	51	5	4.47213595	11.42214869	6.95001274	69500	0	e9f617c64058c825...	5b80b84d...
	52	5	4.47213595	11.42214869	6.95001274	69500	0	cadbca831bfc246c...	5b80b84d...
	53	4	4.00000000	11.42214869	7.42214869	74221	0	69af5e1b7faf6590...	5b80b84d...
	54	4	4.00000000	11.42214869	7.42214869	74221	0	225a14fa0df43690...	5b80b84d...
cmpink!20	55	5	4.47213595	11.42214869	6.95001274	69500	0	7b0fcfd7c9a43d24...	5b80b84d...
	56	5	4.47213595	11.42214869	6.95001274	69500	0	bb2f4f826c449d94...	5b80b84d...
	57	5	4.47213595	11.42214869	6.95001274	69500	0	009bf5d60ac0e584...	5b80b84d...
	58	6	4.89897949	11.42214869	6.52316920	65231	0	3523931c2e378044...	5b80b84d...
	59	5	4.47213595	11.42214869	6.95001274	69500	0	9e8d3b15e51eccb6...	5b80b84d...
	60	7	5.29150262	11.42214869	6.13064607	61306	0	354720a29c74a6f1...	5b80b84d...
	61	4	4.00000000	11.42214869	7.42214869	74221	0	af2f51ef4d5212cf...	5b80b84d...
	62	7	5.29150262	11.42214869	6.13064607	61306	0	398f62e5f9b4f9f4...	5b80b84d...
	63	5	4.47213595	11.42214869	6.95001274	69500	0	23e3aa16071fdfd0...	5b80b84d...
	64	3	3.46410162	11.42214869	7.95804707	79580	1	8b710e72926e75e2...	5b80b84d...
cmpink!20	65	5	4.47213595	11.42214869	6.95001274	69500	0	2fab64e9379b8818...	5b80b84d...
	66	9	6.00000000	11.42214869	5.42214869	54221	0	bda161a697547eb4...	5b80b84d...
	67	5	4.47213595	11.42214869	6.95001274	69500	0	04ff5de7b255089f...	5b80b84d...
	68	7	5.29150262	11.42214869	6.13064607	61306	0	71df72e713a4860e...	5b80b84d...
	69	7	5.29150262	11.42214869	6.13064607	61306	0	eedee0d42ce555a6...	5b80b84d...

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M9-All GRH Table (continued)

N	g	$2\sqrt{g}$	C_{S_4}	margin	VALOR	CM	Cert_SHA (first 16)	parent_M7
70	9	6.00000000	11.42214869	5.42214869	54221	0	dfa3b88fbb45beb2...	5b80b84d...
71	6	4.89897949	11.42214869	6.52316920	65231	0	b4bf6122e34ec057...	5b80b84d...
72	5	4.47213595	11.42214869	6.95001274	69500	0	1186383fc96ea4f1...	5b80b84d...
73	5	4.47213595	11.42214869	6.95001274	69500	0	9c449c50d9587d22...	5b80b84d...
74	8	5.65685425	11.42214869	5.76529444	57652	0	3047eda97a8a696b...	5b80b84d...
75	5	4.47213595	11.42214869	6.95001274	69500	0	5327c22808780697...	5b80b84d...
76	8	5.65685425	11.42214869	5.76529444	57652	0	36fb23cfe3ac0194...	5b80b84d...
77	7	5.29150262	11.42214869	6.13064607	61306	0	a4808274426f5a71...	5b80b84d...
78	11	6.63324958	11.42214869	4.78889911	47888	0	60a769950865b284...	5b80b84d...
79	6	4.89897949	11.42214869	6.52316920	65231	0	009d27cb3712ad33...	5b80b84d...
80	7	5.29150262	11.42214869	6.13064607	61306	0	12cc2b60ebc73fd7...	5b80b84d...
81	4	4.00000000	11.42214869	7.42214869	74221	0	2148f09af399c8a2...	5b80b84d...
82	9	6.00000000	11.42214869	5.42214869	54221	0	495138c1a235204a...	5b80b84d...
83	7	5.29150262	11.42214869	6.13064607	61306	0	7893703842cc75c4...	5b80b84d...
84	11	6.63324958	11.42214869	4.78889911	47888	0	43641ca05365766d...	5b80b84d...
85	7	5.29150262	11.42214869	6.13064607	61306	0	2e888dd13c5fffd45...	5b80b84d...
86	10	6.32455532	11.42214869	5.09759337	50975	0	0e00ef97c4fec921...	5b80b84d...
87	9	6.00000000	11.42214869	5.42214869	54221	0	a74bdf2e13eae3a...	5b80b84d...
88	9	6.00000000	11.42214869	5.42214869	54221	0	5a97e2b31eb1c114...	5b80b84d...
89	7	5.29150262	11.42214869	6.13064607	61306	0	d8bb9099fb243233...	5b80b84d...
90	11	6.63324958	11.42214869	4.78889911	47888	0	c0901624e5d1e058...	5b80b84d...
91	7	5.29150262	11.42214869	6.13064607	61306	0	003463290bc335a4...	5b80b84d...
92	10	6.32455532	11.42214869	5.09759337	50975	0	c9a62d95955a4caa...	5b80b84d...
93	9	6.00000000	11.42214869	5.42214869	54221	0	3634e2f53277a220...	5b80b84d...
94	11	6.63324958	11.42214869	4.78889911	47888	0	e54667cd3add6a29...	5b80b84d...
95	9	6.00000000	11.42214869	5.42214869	54221	0	dd970d4633ef71fb...	5b80b84d...
96	9	6.00000000	11.42214869	5.42214869	54221	0	e08c42d83fa349bd...	5b80b84d...
97	7	5.29150262	11.42214869	6.13064607	61306	0	d36107518a1f1785...	5b80b84d...
98	7	5.29150262	11.42214869	6.13064607	61306	0	8538747ccd2f86dc...	5b80b84d...
99	9	6.00000000	11.42214869	5.42214869	54221	0	43d76e5dc8e4db23...	5b80b84d...
100	7	5.29150262	11.42214869	6.13064607	61306	0	61be40be04557206...	5b80b84d...
101	8	5.65685425	11.42214869	5.76529444	57652	0	8ababd03058a197e...	5b80b84d...
102	15	7.74596669	11.42214869	3.67618200	36761	0	831b51f661b8798b...	5b80b84d...
103	8	5.65685425	11.42214869	5.76529444	57652	0	867be741bae8a25f...	5b80b84d...
104	11	6.63324958	11.42214869	4.78889911	47888	0	f78589a81d9e676e...	5b80b84d...

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M9-All GRH Table (continued)

	N	g	$2\sqrt{g}$	C_{S_4}	margin	VALOR	CM	Cert_SHA (first 16)	parent_M7
	105	13	7.21110255	11.42214869	4.21104614	42110	0	3379327d8c5b56e3...	5b80b84d...
	106	12	6.92820323	11.42214869	4.49394546	44939	0	6b4303ff20207f00...	5b80b84d...
	107	9	6.00000000	11.42214869	5.42214869	54221	0	21f429815f5cbd8b...	5b80b84d...
	108	10	6.32455532	11.42214869	5.09759337	50975	0	325747395012b6d9...	5b80b84d...
	109	8	5.65685425	11.42214869	5.76529444	57652	0	e88c4ddb3ed2ceda...	5b80b84d...
	110	15	7.74596669	11.42214869	3.67618200	36761	0	4f6e458ca7bb458d...	5b80b84d...
	111	11	6.63324958	11.42214869	4.78889911	47888	0	b9762c3a0c723d2c...	5b80b84d...
	112	11	6.63324958	11.42214869	4.78889911	47888	0	f165319f08507c69...	5b80b84d...
	113	9	6.00000000	11.42214869	5.42214869	54221	0	796d41cbcf75f087...	5b80b84d...
	114	17	8.24621125	11.42214869	3.17593744	31759	0	e95788bd0856a6f8...	5b80b84d...
	115	11	6.63324958	11.42214869	4.78889911	47888	0	419cd627a4d179e0...	5b80b84d...
	116	13	7.21110255	11.42214869	4.21104614	42110	0	d652f7cd6647ad31...	5b80b84d...
	117	11	6.63324958	11.42214869	4.78889911	47888	0	212b2f204cda32c3...	5b80b84d...
	118	14	7.48331477	11.42214869	3.93883392	39388	0	865619676999e785...	5b80b84d...
	119	11	6.63324958	11.42214869	4.78889911	47888	0	d28f7e244e62af6f...	5b80b84d...
	120	17	8.24621125	11.42214869	3.17593744	31759	0	aff5758140dd6dad...	5b80b84d...
cmpink!20	121	6	4.89897949	11.42214869	6.52316920	65231	1	68bb883e455cb3aa...	5b80b84d...
	122	14	7.48331477	11.42214869	3.93883392	39388	0	3712e24da8cfe53f...	5b80b84d...
	123	13	7.21110255	11.42214869	4.21104614	42110	0	459fd8dc8a3bb847...	5b80b84d...
	124	14	7.48331477	11.42214869	3.93883392	39388	0	370a646fbecc72b7...	5b80b84d...
	125	8	5.65685425	11.42214869	5.76529444	57652	0	3f5d01b21d6423ef...	5b80b84d...
	126	17	8.24621125	11.42214869	3.17593744	31759	0	9287ce9cd31fb21e...	5b80b84d...
	127	10	6.32455532	11.42214869	5.09759337	50975	0	ef464f9160da8891...	5b80b84d...
	128	9	6.00000000	11.42214869	5.42214869	54221	0	442245bef14a26cf...	5b80b84d...
	129	13	7.21110255	11.42214869	4.21104614	42110	0	e00c72f9c9060006...	5b80b84d...
	130	17	8.24621125	11.42214869	3.17593744	31759	0	1d85304245f1eae9...	5b80b84d...
	131	11	6.63324958	11.42214869	4.78889911	47888	0	922761b56a35df1e...	5b80b84d...
	132	19	8.71779789	11.42214869	2.70435080	27043	0	19bdaf264dc70aaa...	5b80b84d...
	133	11	6.63324958	11.42214869	4.78889911	47888	0	7fc1a50e125230a4...	5b80b84d...
	134	16	8.00000000	11.42214869	3.42214869	34221	0	a8c143ed588cdbba...	5b80b84d...
	135	13	7.21110255	11.42214869	4.21104614	42110	0	cafa6edc3871b40d...	5b80b84d...
	136	15	7.74596669	11.42214869	3.67618200	36761	0	fe804d14d3f2888d...	5b80b84d...
	137	11	6.63324958	11.42214869	4.78889911	47888	0	4a3494461f9bd30e...	5b80b84d...
	138	21	9.16515139	11.42214869	2.25699730	22569	0	ea06864280a5a867...	5b80b84d...
	139	11	6.63324958	11.42214869	4.78889911	47888	0	3b6b47dd3cdeef6...	5b80b84d...

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M9-All GRH Table (continued)

	N	g	$2\sqrt{g}$	C_{S_4}	margin	VALOR	CM	Cert_SHA (first 16)	parent_M7
	140	19	8.71779789	11.42214869	2.70435080	27043	0	e3651b32d8655f16...	5b80b84d...
	141	15	7.74596669	11.42214869	3.67618200	36761	0	73b3ca7fa1659e71...	5b80b84d...
	142	17	8.24621125	11.42214869	3.17593744	31759	0	17e7ce6332236886...	5b80b84d...
	143	13	7.21110255	11.42214869	4.21104614	42110	0	624b93f7d4687b81...	5b80b84d...
cmpink!20	144	13	7.21110255	11.42214869	4.21104614	42110	1	effc4c42fbbd8bce...	5b80b84d...
	145	13	7.21110255	11.42214869	4.21104614	42110	0	d71ab2f279146211...	5b80b84d...
	146	17	8.24621125	11.42214869	3.17593744	31759	0	7e14e8bb44d9589b...	5b80b84d...
	147	11	6.63324958	11.42214869	4.78889911	47888	0	06db068576b0d693...	5b80b84d...
	148	17	8.24621125	11.42214869	3.17593744	31759	0	2fd5c6d72f78569b...	5b80b84d...
	149	12	6.92820323	11.42214869	4.49394546	44939	0	69a0b3900856a4df...	5b80b84d...
	150	19	8.71779789	11.42214869	2.70435080	27043	0	354c092db75635b4...	5b80b84d...
	151	12	6.92820323	11.42214869	4.49394546	44939	0	ce4fa127fa3037ed...	5b80b84d...
	152	17	8.24621125	11.42214869	3.17593744	31759	0	234736d162156b7d...	5b80b84d...
	153	15	7.74596669	11.42214869	3.67618200	36761	0	de52af44313cdce8...	5b80b84d...
	154	21	9.16515139	11.42214869	2.25699730	22569	0	29c0162bf811f8de...	5b80b84d...
	155	15	7.74596669	11.42214869	3.67618200	36761	0	274f998b93a0597c...	5b80b84d...
	156	23	9.59166305	11.42214869	1.83048564	18304	0	81a4e2effd6e94ec...	5b80b84d...
	157	12	6.92820323	11.42214869	4.49394546	44939	0	9dc5ab6e317f8d3f...	5b80b84d...
	158	19	8.71779789	11.42214869	2.70435080	27043	0	6892a44bf2822a92...	5b80b84d...
	159	17	8.24621125	11.42214869	3.17593744	31759	0	5d01de2632f3309f...	5b80b84d...
	160	17	8.24621125	11.42214869	3.17593744	31759	0	3d528bd1805ef604...	5b80b84d...
	161	15	7.74596669	11.42214869	3.67618200	36761	0	01c8bb15705a242b...	5b80b84d...
	162	16	8.00000000	11.42214869	3.42214869	34221	0	36b3472315eba0a5...	5b80b84d...
	163	13	7.21110255	11.42214869	4.21104614	42110	0	99e09b3f02b6a103...	5b80b84d...
	164	19	8.71779789	11.42214869	2.70435080	27043	0	8c618ca0909c98ff...	5b80b84d...
	165	21	9.16515139	11.42214869	2.25699730	22569	0	473dbd6e853a0a85...	5b80b84d...
	166	20	8.94427191	11.42214869	2.47787678	24778	0	c9320b0a83a2bbde...	5b80b84d...
	167	14	7.48331477	11.42214869	3.93883392	39388	0	c7ff5ca2c9ac7974...	5b80b84d...
	168	25	10.00000000	11.42214869	1.42214869	14221	0	193bedf01ecc84b9...	5b80b84d...
cmpink!20	169	8	5.65685425	11.42214869	5.76529444	57652	1	3dc00a5498efa2e5...	5b80b84d...
	170	23	9.59166305	11.42214869	1.83048564	18304	0	1c2e41b9e7f37ea1...	5b80b84d...
	171	17	8.24621125	11.42214869	3.17593744	31759	0	68389d664c3538e2...	5b80b84d...
	172	20	8.94427191	11.42214869	2.47787678	24778	0	d0940d73770a46df...	5b80b84d...
	173	14	7.48331477	11.42214869	3.93883392	39388	0	5392dbd6bef421bb...	5b80b84d...
	174	27	10.39230485	11.42214869	1.02984384	10298	0	ebcc7422bc1fb674...	5b80b84d...

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M9-All GRH Table (continued)

	N	g	$2\sqrt{g}$	C_{S_4}	margin	VALOR	CM	Cert_SHA (first 16)	parent_M7
	175	15	7.74596669	11.42214869	3.67618200	36761	0	9734819fe2cfe506...	5b80b84d...
	176	19	8.71779789	11.42214869	2.70435080	27043	0	bc829a58f5e22fda...	5b80b84d...
	177	19	8.71779789	11.42214869	2.70435080	27043	0	786b36d24dd0fab0...	5b80b84d...
	178	21	9.16515139	11.42214869	2.25699730	22569	0	4933f8efbc8e6e60...	5b80b84d...
	179	15	7.74596669	11.42214869	3.67618200	36761	0	9c33461d45ed84e0...	5b80b84d...
	180	25	10.00000000	11.42214869	1.42214869	14221	0	cabe7691abdc58e7...	5b80b84d...
	181	14	7.48331477	11.42214869	3.93883392	39388	0	c07be4aaab18b666...	5b80b84d...
	182	25	10.00000000	11.42214869	1.42214869	14221	0	f3fa6e053bec5091...	5b80b84d...
	183	19	8.71779789	11.42214869	2.70435080	27043	0	7536fede4961c877...	5b80b84d...
	184	21	9.16515139	11.42214869	2.25699730	22569	0	d1219da5da311d56...	5b80b84d...
	185	17	8.24621125	11.42214869	3.17593744	31759	0	fe82a479e74b9db9...	5b80b84d...
	186	29	10.77032961	11.42214869	0.65181908	6518	0	59fdc55486928fd6...	5b80b84d...
	187	17	8.24621125	11.42214869	3.17593744	31759	0	9c23740306c5948a...	5b80b84d...
	188	22	9.38083152	11.42214869	2.04131717	20413	0	c79ba557c7105f70...	5b80b84d...
	189	19	8.71779789	11.42214869	2.70435080	27043	0	dcd03dfcad81dbcc...	5b80b84d...
	190	27	10.39230485	11.42214869	1.02984384	10298	0	efb11150be5eb20b...	5b80b84d...
	191	16	8.00000000	11.42214869	3.42214869	34221	0	1298d166640339f5...	5b80b84d...
	192	21	9.16515139	11.42214869	2.25699730	22569	0	025be425e9dd6720...	5b80b84d...
	193	15	7.74596669	11.42214869	3.67618200	36761	0	d5aa8234fa0fe1a9...	5b80b84d...
	194	23	9.59166305	11.42214869	1.83048564	18304	0	9d2c11f4cd530a14...	5b80b84d...
	195	25	10.00000000	11.42214869	1.42214869	14221	0	eb46970290fa455a...	5b80b84d...
cmpink!20	196	17	8.24621125	11.42214869	3.17593744	31759	1	ec818827598d5351...	5b80b84d...
	197	16	8.00000000	11.42214869	3.42214869	34221	0	55a01a7ebabb83d5...	5b80b84d...
	198	29	10.77032961	11.42214869	0.65181908	6518	0	f7cc3eb060c25cc5...	5b80b84d...
	199	16	8.00000000	11.42214869	3.42214869	34221	0	40d04588e393d5f0...	5b80b84d...
	200	19	8.71779789	11.42214869	2.70435080	27043	0	22122cbcf482585a...	5b80b84d...
	201	21	9.16515139	11.42214869	2.25699730	22569	0	922eaf40fce057b3...	5b80b84d...
	202	24	9.79795897	11.42214869	1.62418972	16241	0	5b2cdfdd57442ebc...	5b80b84d...
	203	19	8.71779789	11.42214869	2.70435080	27043	0	a135273b87e3ac0c...	5b80b84d...
	204	31	11.13552873	11.42214869	0.28661996	2866	0	fed5ed2d11eb010e...	5b80b84d...
	205	19	8.71779789	11.42214869	2.70435080	27043	0	119d888e73ff341e...	5b80b84d...
	206	25	10.00000000	11.42214869	1.42214869	14221	0	f8edfc425b55fbca...	5b80b84d...
	207	21	9.16515139	11.42214869	2.25699730	22569	0	5463d02a2fac4085...	5b80b84d...
	208	23	9.59166305	11.42214869	1.83048564	18304	0	bc963a7a4df16de4...	5b80b84d...
	209	19	8.71779789	11.42214869	2.70435080	27043	0	a7b147b109743bb0...	5b80b84d...

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M9-All GRH Table (continued)

N	g	$2\sqrt{g}$	C_{S_4}	margin	VALOR	CM	Cert_SHA (first 16)	parent_M7
211	17	8.24621125	11.42214869	3.17593744	31759	0	206d9a24a2edfeb1...	5b80b84d...
212	25	10.00000000	11.42214869	1.42214869	14221	0	7d387b75d45e1762...	5b80b84d...
213	23	9.59166305	11.42214869	1.83048564	18304	0	30565b3dc6dc3f4f...	5b80b84d...
214	26	10.19803903	11.42214869	1.22410966	12241	0	461bfafa9bfca408...	5b80b84d...
215	21	9.16515139	11.42214869	2.25699730	22569	0	3816adf5a84f0016...	5b80b84d...
216	25	10.00000000	11.42214869	1.42214869	14221	0	0de2db18ce8ced9c...	5b80b84d...
217	19	8.71779789	11.42214869	2.70435080	27043	0	6fff8ed5cef37e99...	5b80b84d...
218	26	10.19803903	11.42214869	1.22410966	12241	0	9d4d4745035d3052...	5b80b84d...
219	23	9.59166305	11.42214869	1.83048564	18304	0	baf398adf96b5cce...	5b80b84d...
220	31	11.13552873	11.42214869	0.28661996	2866	0	0e35cc6eb455324a...	5b80b84d...
221	19	8.71779789	11.42214869	2.70435080	27043	0	e3ab362e39daa993...	5b80b84d...
223	18	8.48528137	11.42214869	2.93686732	29368	0	104e395249a41778...	5b80b84d...
224	25	10.00000000	11.42214869	1.42214869	14221	0	73da62f7a19bd111...	5b80b84d...
225	19	8.71779789	11.42214869	2.70435080	27043	0	aa3a38fa053118c3...	5b80b84d...
226	27	10.39230485	11.42214869	1.02984384	10298	0	42b3a3544b7b76cd...	5b80b84d...
227	19	8.71779789	11.42214869	2.70435080	27043	0	439f3a19415abbc3...	5b80b84d...
229	18	8.48528137	11.42214869	2.93686732	29368	0	533ad8b8474b2f10...	5b80b84d...
231	29	10.77032961	11.42214869	0.65181908	6518	0	b710b2236912fcdc...	5b80b84d...
232	27	10.39230485	11.42214869	1.02984384	10298	0	85f8d54f45ba7386...	5b80b84d...
233	19	8.71779789	11.42214869	2.70435080	27043	0	b09c458b55fd3393...	5b80b84d...
235	23	9.59166305	11.42214869	1.83048564	18304	0	bf4744a2077673cc...	5b80b84d...
236	28	10.58300524	11.42214869	0.83914345	8391	0	35d6ea1e641fc65a...	5b80b84d...
237	25	10.00000000	11.42214869	1.42214869	14221	0	3c6aec9592391ca4...	5b80b84d...
239	20	8.94427191	11.42214869	2.47787678	24778	0	953632b5720e26a3...	5b80b84d...
241	19	8.71779789	11.42214869	2.70435080	27043	0	12f6fca8013251a4...	5b80b84d...
242	22	9.38083152	11.42214869	2.04131717	20413	0	4afe1b950ca786df...	5b80b84d...
243	19	8.71779789	11.42214869	2.70435080	27043	0	1f18b09228e45b8e...	5b80b84d...
244	29	10.77032961	11.42214869	0.65181908	6518	0	6bdce0552d394b9a...	5b80b84d...
245	21	9.16515139	11.42214869	2.25699730	22569	0	7a9c0415eeb47bf6...	5b80b84d...
247	21	9.16515139	11.42214869	2.25699730	22569	0	fa033bbd4354059c...	5b80b84d...
248	29	10.77032961	11.42214869	0.65181908	6518	0	4f254467eadce052...	5b80b84d...
249	27	10.39230485	11.42214869	1.02984384	10298	0	0a112758cf3470c8...	5b80b84d...
250	28	10.58300524	11.42214869	0.83914345	8391	0	4cabe8ca0efd3e37...	5b80b84d...
251	21	9.16515139	11.42214869	2.25699730	22569	0	2d2ef73481be60a0...	5b80b84d...
253	23	9.59166305	11.42214869	1.83048564	18304	0	d20f617eeb685c8c...	5b80b84d...

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M9-All GRH Table (continued)

	N	g	$2\sqrt{g}$	C_{S_4}	margin	VALOR	CM	Cert_SHA (first 16)	parent_M7
cmpink!20	254	31	11.13552873	11.42214869	0.28661996	2866	0	86693f5422e57207...	5b80b84d...
	256	21	9.16515139	11.42214869	2.25699730	22569	1	bea770c5946940d8...	5b80b84d...
	257	21	9.16515139	11.42214869	2.25699730	22569	0	4003385c5e6a3fef...	5b80b84d...
	259	23	9.59166305	11.42214869	1.83048564	18304	0	dab3ebbf6e61da2...	5b80b84d...
	261	27	10.39230485	11.42214869	1.02984384	10298	0	b8bee3f794b7c8b6...	5b80b84d...
	262	32	11.31370850	11.42214869	0.10844019	1084	0	a74652e630cd33e5...	5b80b84d...
	263	22	9.38083152	11.42214869	2.04131717	20413	0	1d048e7e3e37a11c...	5b80b84d...
	265	25	10.00000000	11.42214869	1.42214869	14221	0	111987ee03ef093b...	5b80b84d...
	267	29	10.77032961	11.42214869	0.65181908	6518	0	e9e43ab324fbf6a5...	5b80b84d...
	268	32	11.31370850	11.42214869	0.10844019	1084	0	4ff455edebe4a395...	5b80b84d...
	269	22	9.38083152	11.42214869	2.04131717	20413	0	ef4107e52f2cbc3d...	5b80b84d...
	271	22	9.38083152	11.42214869	2.04131717	20413	0	a3f57429bb0983ad...	5b80b84d...
	272	31	11.13552873	11.42214869	0.28661996	2866	0	5054c40c26b6709f...	5b80b84d...
	275	25	10.00000000	11.42214869	1.42214869	14221	0	f8ce7c1d011e4cae...	5b80b84d...
	277	22	9.38083152	11.42214869	2.04131717	20413	0	9b7affcb9ae248f7...	5b80b84d...
	279	29	10.77032961	11.42214869	0.65181908	6518	0	8ee50ebd2ff3432f...	5b80b84d...
	281	23	9.59166305	11.42214869	1.83048564	18304	0	bd291eca0f89c675...	5b80b84d...
	283	23	9.59166305	11.42214869	1.83048564	18304	0	6532e7304be06696...	5b80b84d...
	287	27	10.39230485	11.42214869	1.02984384	10298	0	b11603b1508905b8...	5b80b84d...
	289	17	8.24621125	11.42214869	3.17593744	31759	0	6172bbdabb2a9f24...	5b80b84d...
	291	31	11.13552873	11.42214869	0.28661996	2866	0	f899df8a39659c6d...	5b80b84d...
	293	24	9.79795897	11.42214869	1.62418972	16241	0	ee3c1600ca246450...	5b80b84d...
	295	29	10.77032961	11.42214869	0.65181908	6518	0	49de4c516540b23f...	5b80b84d...
	297	31	11.13552873	11.42214869	0.28661996	2866	0	c47afacd8d1a38f7...	5b80b84d...
	299	27	10.39230485	11.42214869	1.02984384	10298	0	980e650603adabdf...	5b80b84d...
	301	27	10.39230485	11.42214869	1.02984384	10298	0	fe9cd7f0d39e4be1...	5b80b84d...
	305	29	10.77032961	11.42214869	0.65181908	6518	0	372290624250d225...	5b80b84d...
	307	25	10.00000000	11.42214869	1.42214869	14221	0	332249d1e374af58...	5b80b84d...
	311	26	10.19803903	11.42214869	1.22410966	12241	0	fad4e56030810ea6...	5b80b84d...
	313	25	10.00000000	11.42214869	1.42214869	14221	0	483d39ef1abed84c...	5b80b84d...
	317	26	10.19803903	11.42214869	1.22410966	12241	0	06004b6d07f2bcbd...	5b80b84d...
	319	29	10.77032961	11.42214869	0.65181908	6518	0	1288da8b732f0d85...	5b80b84d...
	323	29	10.77032961	11.42214869	0.65181908	6518	0	a5ee849eedee1a48...	5b80b84d...
	325	29	10.77032961	11.42214869	0.65181908	6518	0	a16a4368490233fe...	5b80b84d...
	329	31	11.13552873	11.42214869	0.28661996	2866	0	4c3cb0e07438c91f...	5b80b84d...

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M9-All GRH Table (continued)

	N	g	$2\sqrt{g}$	C_{S_4}	margin	VALOR	CM	Cert_SHA (first 16)	parent_M7
	331	27	10.39230485	11.42214869	1.02984384	10298	0	f9468daa84cc29e0...	5b80b84d...
	337	27	10.39230485	11.42214869	1.02984384	10298	0	94a97ecea95b4afa...	5b80b84d...
cmpink!20	338	32	11.31370850	11.42214869	0.10844019	1084	1	88c20a221b45f9c2...	5b80b84d...
	341	31	11.13552873	11.42214869	0.28661996	2866	0	3f4ea6adc8275783...	5b80b84d...
	343	26	10.19803903	11.42214869	1.22410966	12241	0	0617eefac42f0237...	5b80b84d...
	347	29	10.77032961	11.42214869	0.65181908	6518	0	6934506db195f5e7...	5b80b84d...
	349	28	10.58300524	11.42214869	0.83914345	8391	0	2aa2a8bb00e08ffa...	5b80b84d...
	353	29	10.77032961	11.42214869	0.65181908	6518	0	7b70e6d32837a5af...	5b80b84d...
	359	30	10.95445115	11.42214869	0.46769754	4676	0	34e1490a6eacfea2...	5b80b84d...
cmpink!20	361	22	9.38083152	11.42214869	2.04131717	20413	1	e68da70a577dd15d...	5b80b84d...
	367	30	10.95445115	11.42214869	0.46769754	4676	0	febbdd3837f5d124...	5b80b84d...
	373	30	10.95445115	11.42214869	0.46769754	4676	0	f04ee08eda0d64fe...	5b80b84d...
	379	31	11.13552873	11.42214869	0.28661996	2866	0	1ed0444bc0278cdc...	5b80b84d...
	383	32	11.31370850	11.42214869	0.10844019	1084	0	e1b9e21eade8d5c6...	5b80b84d...
	389	32	11.31370850	11.42214869	0.10844019	1084	0	fb2eaa8a8cf7b35b...	5b80b84d...
	397	32	11.31370850	11.42214869	0.10844019	1084	0	b53ec7e7c70f15a4...	5b80b84d...

Summary:

- TOTAL_ROWS: 280
- MIN_VALOR: 1084 (at N = 262, 268, 338, 383, 389, 397; g = 32)
- MIN_MARGIN: 0.10844019
- ALL_VALOR_POSITIVE: **true**
- M24_READY: **true**

Certified – Opera Numerorum / Battle Plan v1.6 – David Fox – May 2026